Hospital AI Agent

System Requirements Specification (SRS)

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1. Introduction

The Hospital AI Agent is an AI-powered smart hospital management system designed to connect hospital administrators, doctors, nurses, receptionists, and patients through one integrated platform.

The system aims to automate hospital management processes and provide intelligent assistance through an AI Agent.

The system will manage users, appointments, schedules, patient information, notifications, reports, and intelligent recommendations.

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2. Functional Requirements

Functional requirements describe what the system must be able to do.

2.1 User Authentication and Authorization

FR-01: User Registration

The system shall allow authorized users to register new accounts.

The system shall support the following user roles:

- Administrator.
- Doctor.
- Nurse.
- Receptionist.
- Patient.

FR-02: User Login

The system shall allow registered users to log into the system using their credentials.

FR-03: Role-Based Access Control

The system shall provide different permissions depending on the user's role.

Each user shall only be able to access the features and information authorized for their role.

2.2 Hospital Administration

FR-04: Dashboard

The system shall provide an administrative dashboard that displays:

- Total number of doctors.
- Total number of nurses.
- Total number of patients.
- Number of daily appointments.
- Number of completed appointments.
- Daily revenue.
- Monthly revenue.
- Hospital performance statistics.

FR-05: Doctor Management

The administrator shall be able to:

- Add a doctor.
- Edit doctor information.
- Delete a doctor.
- View doctor schedules.
- View doctor appointments.
- View the number of completed appointments.

- Monitor doctor workload.

FR-06: Nurse Management

The administrator shall be able to:

- Add nurses.
- Edit nurse information.
- Assign nurses to departments.
- Assign nurses to doctors.
- Create and manage shifts.
- View nurse schedules.

2.3 Patient Management

FR-07: Patient Registration

The system shall allow authorized users to register new patients.

The system shall store:

- Patient name.
- Contact information.
- Date of birth or age.
- Medical history.
- Previous visits.

FR-08: Patient Search

The system shall allow authorized users to search for patients using:

- Patient name.
- Patient ID.
- Phone number.

FR-09: Medical History

The system shall store and allow authorized healthcare professionals to access patient medical history.

Access to medical information shall be controlled based on user permissions.

2.4 Doctor Management

FR-10: Doctor Schedule

The system shall allow doctors and administrators to view doctor schedules.

The schedule shall include:

- Working days.
- Working hours.
- Appointments.
- Procedures.
- Special sessions.

FR-11: Appointment Statistics

The system shall calculate:

- Number of booked appointments.
- Number of completed appointments.
- Number of cancelled appointments.

FR-12: Doctor Income

The system may calculate doctor income based on completed appointments and configured payment rules.

2.5 Appointment Management

FR-13: Book Appointment

The system shall allow patients or authorized staff to book appointments.

The booking process shall include:

1. Selecting a medical specialty.
2. Selecting a doctor.
3. Viewing available appointment slots.
4. Selecting an appointment date and time.
5. Confirming the appointment.

FR-14: Modify Appointment

The system shall allow authorized users to modify existing appointments.

FR-15: Cancel Appointment

The system shall allow authorized users to cancel appointments.

FR-16: Appointment Status

The system shall manage appointment statuses, including:

- Booked.
- Completed.
- Cancelled.

2.6 Notification System

FR-17: Appointment Reminders

The system shall send appointment reminders to patients.

Notifications may be sent:

- 24 hours before the appointment.

- 1 hour before the appointment.

FR-18: Doctor Notifications

The system shall notify doctors about:

- Upcoming working schedules.
- Upcoming appointments.

FR-19: Nurse Notifications

The system shall notify nurses before the beginning of their shifts.

FR-20: Administrative Alerts

The system shall notify administrators when:

- A department becomes overcrowded.
- A doctor has an unusually high workload.
- There is a shortage of staff in a department.

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3. AI Agent Requirements

FR-21: Natural Language Interaction

The AI Agent shall allow users to interact with the system using natural language.

Example:

“I want to book an appointment.”

FR-22: Doctor Recommendation

The AI Agent shall assist users in finding an appropriate medical specialty or doctor based on the user's stated needs.

The AI Agent shall not provide a final medical diagnosis.

FR-23: Appointment Recommendation

The AI Agent shall:

- Search for available doctors.
- Check available appointment slots.
- Suggest suitable appointment times.

FR-24: Appointment Assistance

The AI Agent shall assist authorized users with appointment booking through approved system tools.

FR-25: Hospital Information Assistance

The AI Agent shall answer general hospital-related questions.

Examples include:

- Available doctors.
- Medical specialties.
- Working hours.
- Appointment availability.

FR-26: Patient History Summary

The AI Agent may summarize authorized patient medical history for healthcare professionals.

The summary shall only be available to authorized users.

FR-27: Workload Analysis

The AI Agent shall analyze hospital operational data to identify:

- Overloaded doctors.
- Busy departments.
- High-demand periods.

FR-28: Intelligent Recommendations

The AI Agent may provide recommendations such as:

- Alternative appointment times.
- Less crowded doctors.
- Improved nurse distribution.

Recommendations shall be presented as decision support and remain subject to human approval.

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4. Reporting Requirements

FR-29: Daily Reports

The system shall generate reports about:

- Daily patients.
- Daily appointments.
- Completed appointments.
- Daily revenue.

FR-30: Monthly Reports

The system shall generate reports about:

- Monthly patients.
- Monthly appointments.
- Monthly revenue.

- Department performance.

FR-31: Doctor Reports

The system shall generate reports containing:

- Number of appointments.
- Completed appointments.
- Cancelled appointments.
- Doctor workload.
- Doctor income.

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5. Non-Functional Requirements

Non-functional requirements describe how the system should perform.

NFR-01: Security

The system shall protect user accounts and sensitive information.

Passwords shall be securely stored.

Access to sensitive data shall be controlled based on user roles.

NFR-02: Privacy

Patient information shall only be accessible to authorized users.

The system shall protect sensitive medical data.

NFR-03: Performance

The system should provide normal responses to common user requests within an acceptable time.

The system should support multiple users.

NFR-04: Usability

The system shall provide a simple and user-friendly interface.

The system should be easy to use for:

- Administrators.
- Doctors.
- Nurses.
- Receptionists.
- Patients.

NFR-05: Reliability

The system should maintain data consistency and reduce the possibility of data loss.

NFR-06: Scalability

The system should be designed so that additional hospital departments and users can be added in the future.

NFR-07: Maintainability

The system should use a modular architecture to make future updates and maintenance easier.

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6. User Roles and Permissions

Administrator

The Administrator can:

- Manage doctors.
- Manage nurses.
- Manage hospital departments.

- View reports.
- View statistics.
- Monitor workload.
- Manage system users.

Doctor

The Doctor can:

- View appointments.
- View authorized patient information.
- Manage appointment status.
- Add medical notes.
- Add prescriptions.
- View schedules.

Nurse

The Nurse can:

- View assigned shifts.
- View assigned departments.
- View assigned doctors.
- Confirm shift completion.

Receptionist

The Receptionist can:

- Register patients.
- Search for patients.
- Book appointments.

- Modify appointments.
- Cancel appointments.

Patient

The Patient can:

- View available doctors.
- Book appointments.
- Modify appointments.
- Cancel appointments.
- View appointment information.
- Receive notifications.
- Interact with the AI Agent.

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7. System Constraints

The project will initially focus on a Minimum Viable Product (MVP).

The first version may not include all advanced features.

The AI Agent will have controlled access to the system through predefined tools and APIs.

The AI Agent will not make final medical decisions or diagnoses.

Sensitive patient information will only be available to authorized users.

Notification services such as SMS or WhatsApp may require external APIs and may be implemented in future phases.

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8. MVP Requirements

The first version of the system shall include the following core features.

User Management

- Login.

- Role-based access.

Doctor Management

- Add doctors.
- View doctors.
- Manage doctor information.

Patient Management

- Add patients.
- View patients.
- Search patients.

Appointment Management

- View available doctors.
- Book appointments.
- Modify appointments.
- Cancel appointments.

Dashboard

- Display basic statistics.
- Display daily appointments.

AI Agent

The AI Agent MVP shall support:

- Answering basic hospital questions.
- Searching for doctors.
- Suggesting doctors based on specialty.
- Checking available appointment slots.
- Assisting with appointment booking.

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9. Future Requirements

Future versions may include:

- Prescription analysis.
- Medication reminders.
- Treatment tracking.
- Advanced financial reports.
- Automatic nurse distribution.
- Hospital crowd prediction.
- Voice-based AI Agent.
- Advanced medical record summarization.
- Integration with external healthcare systems.

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10. Summary

The Hospital AI Agent system will combine hospital management features with artificial intelligence.

The system will initially focus on managing users, doctors, patients, appointments, and hospital information.

The AI Agent will provide intelligent assistance by understanding user requests, retrieving authorized information, suggesting appropriate actions, and interacting with the system through predefined tools.

The project will be developed gradually, starting with a Minimum Viable Product and expanding to include advanced AI-powered hospital management features.